DEVELOPMENT OF PROTOTYPE GAME TO SIMULATE TERRORIST ATTACKS IN SINGAPORE FOR STRONGER DEFENCE

Group 7:
Valerie Chen Wen Yun, Tan Ern Min,
Kyson Chua Kyi Sheng, Saw Ze Dong,
Lee Yu Bing Darrius, Liaw Yu Jun

Literature Review

Virtual Reality

Computer-generated environment with realistic scenes

Objectives

Create a realistic VR game to train officers to defend Singapore against terrorism

Overseas Use of VR

AUGGMED

Benefit: allow for trainees from multiple agencies to train simultaneously

- a multi-agency counter terrorism training program
- on-site soldiers: **augmented reality**
  other soldiers: **virtual reality** with collaborative training enabled
- mission: effectively respond to terrorist incident, ensure safety of civilians, and neutralise the threat

Overseas Use of VR

IOM UN MIGRATION

ATLAS

Benefit: can be updated frequently to follow recent attacks

- ATLAS created for UN's IOM
- designed to train civilian staff to respond effectively to life-threatening situations
- commissioned to follow increased hostile activity against aid convoys

Singapore's Context

Virtual Simulations

Areas for Improvement: Can develop VR for soldiers to move about freely and train physically while training virtually

- virtual simulation accessed through computers
- simulate terror attacks and unexpected scenarios
- designed to look like terrorist acts are happening on an actual Singapore street

Problem Statements

High Cost

Large sums of money are needed to simulate real-life situations

Limited Accessibility

Opportunities are limited for army personnel to train

Casualties

Drills can cause both injuries and death

High Cost

Bohemia Interactive Simulations, used by Britain for military simulation

- Vehicle repairs, ammunition and guns
- Escalates to hefty costs, unsustainable
- Areas such as Britain and China has adapted AR and VR training to counter budget strains

The Sar-21's for which its ammo can cost up to 0.36 USD to 1 USD per round for its 5.56 ammunition

TechViz Fusion, a virtual reality simulation company that can render an actual battlefield for a complete pilot training

Limited Accessibility

- High cost → fewer training opportunities
- COVID-19 border restrictions
- Army personnel are unable to train overseas
- Singapore is limited in land resources to train
- AR and VR allows training to be unconstrained by space

Boeing F-15SG Eagle at Mountain Home Air Force Base, Idaho, during Exercise Forging Sabre

Casualties

- Physical training → actual casualties
- Higher risks of injuries
- AR and VR provides a more safer environment to prevent accidents from occurring during military training

US Marines transport a non-ambulatory patient via litter

Proposed Solution

VR Simulations

Simulate terrorist attacks to train the Singapore Army for similar future situations

Develops Flexibility

Different scenarios hone spontaneous decision-making skills

Multipurpose

Scenarios situated on both land and sea; suitable for different specialisations

Proposed Solution

Immersive

Users can directly apply their learning points to a real-life scenario, with a quick and apt response

Cost-effective

Lower maintenance; saves budget in the long run

Encourages International Cooperation

Joint operations are made easier as less resources are expended on transportation and sustenance

VR Demonstration

Development of Ideas

Ideation Process

- VR simulation is more realistic and feasible
  - increased accessibility
  - reduced costs and casualties
- enhanced current simulations by incorporating VR element
  - our VR game works well with the VR headset
  - used CoSpaces and TypeScript API

Development of Ideas

Further Enhancement

- develop more levels (include task of saving civilians)
- introduce more player roles (e.g. different ranks of army personnel, medics)
- introduce AR to simulate actual locations of Singapore
- introduce multiplayer mode

Conclusion

- capitalises on prevalence of VR technology
- realistically simulate real-life scenarios
- crucial applications in military training and defence
- keep Singapore safer and less vulnerable to threats in our society

Thank you!